



IMPACT OF ARTIFICIAL INTELLIGENCE (AI) TOOLS ON STUDENTS' LEARNING AND ACADEMIC INTEGRITY

Submitted by

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This research report is submitted to **Faculty of Education, Tribhuvan University** in a partial fulfillment of requirements for the degree of Bachelor in Information, Communication and Technology Education (BICTE)

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APPROVAL SHEET

This is to certify that the research project entitled "Impact of Artificial Intelligence (AI) Tools on Students' Learning and Academic Integrity" has been prepared and submitted by Abishek Khawas in partial fulfilment of the requirements for the degree of Bachelor of Education in Information Communication Technology (BICTE) under the Faculty of Education.

This research work has been thoroughly reviewed and evaluated by the undersigned. In our professional opinion, the project meets the required academic standards with regard to relevance, depth, and overall quality. We, therefore, grant approval for its submission as part of the undergraduate degree program.

Focal Person: _____

External Examiner: _____

Supervisor: _____

Date: July 2025

DECLARATION

I hereby declare that the research proposal entitled “Impact of Artificial Intelligence (AI) Tools on Students’ Learning and Academic Integrity” is my original work and has been conducted in accordance with the academic principles and ethical standards of research. This proposal has not been submitted previously, either in whole or in part, to any other university or institution for the fulfilment of any academic requirement.

I take full responsibility for the content and authenticity of the information presented in this work.

Abishek Khawas

Date: _____

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Table of Contents

APPROVAL SHEET.....	II
DECLARATION.....	III
ACKNOWLEDGEMENT.....	IV
CHAPTER 1: INTRODUCTION.....	1
1.1 Background of the Study.....	1
1.2 Statement of the Problem.....	1
1.3 Objectives of the Study.....	2
General Objective:.....	2
Specific Objectives:.....	2
1.4 Research Questions.....	2
1.5 Significance of the Study.....	3
1.6 Delimitation and Limitation.....	3
Delimitations:.....	3
Limitations:.....	3
CHAPTER 2: REVIEW OF LITERATURE.....	3
2.1 Theoretical Framework.....	3
2.2 Conceptual Framework.....	4
Independent Variables:.....	4
Dependent Variables:.....	4
Moderating Factors:.....	4
2.3 Review of Related Studies.....	4
CHAPTER 3: RESEARCH METHODOLOGY.....	5
3.1 Research Design.....	5
3.2 Population and Sample.....	5
Sample:.....	6
3.3 Sampling Procedure.....	6
3.4 Data Collection Tools.....	6
3.5 Data Collection Procedure.....	6
3.6 Data Analysis Techniques.....	6

CHAPTER 4: TIME SCHEDULE (GANTT CHART).....	7
CHAPTER 5: REFERENCES.....	7

List of Tables

Table 1: Gantt Chart(Week Timeline).....	7
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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Artificial Intelligence (AI) tools have revolutionised education by offering personalised, on-demand learning support. Tools such as ChatGPT, developed by OpenAI, are widely adopted by students for tasks ranging from drafting essays to solving complex academic problems. These AI platforms provide instant explanations, summaries, and suggestions, making learning more accessible and interactive. AI also facilitates differentiated instruction by adapting to individual learning paces and styles, thereby supporting both struggling and advanced learners.

However, this growing reliance also brings forth ethical concerns regarding academic integrity, originality, and student dependency. The ease with which students can obtain complete answers without deep engagement may undermine critical thinking and problem-solving skills. Additionally, educators face challenges in assessing genuine student understanding and ensuring fair evaluation.

The duality of AI's role in education—serving as both a learning enhancer and a potential enabler of misconduct—necessitates an in-depth exploration into how such tools affect both learning outcomes and students' ethical conduct. Understanding this dynamic is crucial for developing balanced educational policies and guiding responsible AI usage in academic settings.

1.2 Statement of the Problem

Despite the promising educational benefits of AI tools, concerns are emerging about their misuse. Increasing reports of AI-assisted cheating, plagiarism, and reduced critical engagement are alarming educators and institutions. The convenience and speed of AI-generated responses may encourage students to rely on these tools for completing assignments, bypassing essential steps in the learning process such as research, reflection, and critical thinking.

Moreover, there is a growing uncertainty about whether students clearly understand the ethical boundaries of AI usage. Many institutions are yet to establish comprehensive

guidelines or educate students on appropriate and inappropriate use. As a result, confusion about what constitutes acceptable use of AI tools persists, leading to unintentional academic misconduct in some cases.

This research seeks to understand how students are incorporating AI tools into their academic routines—whether these tools are being used to support genuine learning or to shortcut academic responsibilities. It also aims to assess the extent to which AI use enhances comprehension and performance versus fostering cognitive dependency. Furthermore, the study will examine how student use of AI aligns with institutional academic integrity policies, identify gaps in policy and awareness, and propose strategies to promote responsible and ethical integration of AI in education.

1.3 Objectives of the Study

General Objective:

To explore the impact of Artificial Intelligence (AI) tools on students' learning processes and academic integrity through a qualitative approach.

Specific Objectives:

- i. To investigate the various ways students integrate AI tools into their academic tasks such as assignments, research, and exam preparation.
- ii. To understand students' perceptions of how AI tools affect their cognitive development, critical thinking, and overall learning experience.
- iii. To examine ethical concerns arising from AI use in education, particularly in relation to plagiarism, academic dishonesty, and over dependence.
- iv. To propose evidence-based strategies for promoting responsible, ethical, and educationally beneficial use of AI technologies in academic environments.

1.4 Research Questions

- i. How are students integrating AI tools into their academic activities?
- ii. In what ways do students believe these tools affect their learning and thinking skills?
- iii. What are the ethical concerns faced by students and educators regarding AI use?

- iv. What strategies can institutions adopt to foster ethical and effective AI tool usage?

1.5 Significance of the Study

This study is significant as it provides educational stakeholders—including students, educators, administrators, and policymakers—with a grounded understanding of how AI tools are experienced and perceived in academic contexts. The findings will offer valuable insights into the benefits and challenges associated with AI integration in learning, particularly in relation to maintaining academic integrity. These insights can inform the design and implementation of AI literacy programs, institutional policies, and pedagogical strategies that balance innovation with ethical considerations. Ultimately, this research aims to support the responsible and effective integration of AI technologies in curricula, fostering an environment where technology enhances learning without compromising educational standards.

1.6 Delimitation and Limitation

Delimitations:

- i. The study is limited to higher secondary and undergraduate students.
- ii. Focuses only on AI tools related to language processing and content generation (e.g., ChatGPT).

Limitations:

- i. Findings are based on students' and educators' self-reported experiences and perceptions.
- ii. Results may not remain universally applicable as AI technologies evolve rapidly.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Theoretical Framework

The study is grounded in Constructivist Learning Theory, which posits that learners actively construct knowledge through interaction, reflection, and problem-solving (Piaget, 1950; Vygotsky, 1978). AI tools like ChatGPT can support constructivist learning by providing students with instant, interactive feedback. However, excessive dependence on

these tools may limit deeper cognitive engagement and hinder independent knowledge construction (Kasneci et al., 2023).

2.2 Conceptual Framework

Independent Variables:

Use of AI tools (e.g., ChatGPT, Google Bard, Microsoft Bing AI, Jasper AI, Grammarly AI)

Dependent Variables:

- i. Student learning outcomes (e.g., engagement, comprehension)
- ii. Academic integrity (e.g., honesty, originality)

Moderating Factors:

- i. Institutional AI usage policies
- ii. Digital literacy levels
- iii. Students' purpose for AI tool usage

This framework enables the study to explore not only the effects of AI on education but also the conditions under which its use is ethical and beneficial (Cotton et al., 2023; UNESCO, 2023).

2.3 Review of Related Studies

Kasneci et al. (2023) argue that while AI tools enable personalized and inquiry-based learning, they also risk reducing students' critical thinking if overused, potentially leading to cognitive offloading where learners depend too heavily on AI-generated answers rather than engaging deeply with content. Similarly, Cotton et al. (2023) highlight concerns about the misuse of ChatGPT for plagiarism and academic dishonesty, emphasizing the urgent need for institutional oversight and clear guidelines to safeguard academic standards.

UNESCO (2023) recommends the development of AI literacy programs and ethical frameworks within academic environments to prepare both educators and students for responsible AI integration. Their policy guidance stresses that ethical AI use must be

embedded in curricula to balance innovation with integrity. OpenAI (2023) also emphasizes the importance of human supervision and pedagogical strategies to prevent misuse, advocating for AI as an assistive tool rather than a substitute for critical thinking.

Further, Zhang et al. (2024) explored student attitudes towards AI in higher education, finding a dual perspective: while many appreciate AI's ability to enhance learning efficiency and accessibility, there is apprehension about its impact on creativity and independent problem-solving. They stress the importance of scaffolding AI use within instructional design to maximize benefits while minimizing risks.

In a study by Williams and Nguyen (2023), educators reported challenges in detecting AI-generated content, underscoring the need for improved assessment techniques and AI-detection tools. They call for collaboration between academic institutions and technology developers to address these challenges effectively.

Moreover, Bower et al. (2023) emphasize that AI's potential in education extends beyond content generation to include personalized feedback, adaptive assessments, and support for diverse learning needs. However, they caution that without proper ethical guidelines and digital literacy, such benefits may be undermined by misuse or overreliance.

Collectively, these studies underscore the critical need for a balanced, well-guided integration of AI tools in education, combining technological innovation with ethical vigilance and pedagogical adaptation.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This is a qualitative exploratory study aimed at understanding students' experiences, perceptions, and ethical dilemmas associated with AI tools. A phenomenological approach will be used to capture lived experiences.

3.2 Population and Sample

Higher secondary and undergraduate students and educators from selected institutions.

Sample:

- i. Students: 20 participants (via Focus Group Discussions)
- ii. Educators: 8 participants (via In-depth Interviews)

3.3 Sampling Procedure

- i. Students: Purposive sampling to ensure participants who actively use AI tools are selected.
- ii. Educators: Expert sampling to include those experienced in academic assessment and digital education.

3.4 Data Collection Tools

- i. Semi-structured interview guides for one-on-one interviews
- ii. Focus Group Discussion (FGD) protocols for student groups

3.5 Data Collection Procedure

- i. Interviews and FGDs will be conducted either in-person or virtually (Zoom/Google Meet).
- ii. Informed consent will be taken from all participants. Sessions will be audio-recorded and transcribed.

3.6 Data Analysis Techniques

Thematic Analysis using Braun and Clarke's (2006) six-phase approach:

- i. Familiarisation with data
- ii. Coding
- iii. Searching for themes
- iv. Reviewing themes
- v. Defining and naming themes
- vi. Writing up findings

CHAPTER 4: TIME SCHEDULE (GANTT CHART)

Table 1: Gantt Chart(Week Timeline)

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Proposal Approval	✓							
Literature Review		✓	✓					
Tool Development			✓					
Data Collection				✓	✓			
Data Analysis						✓	✓	
Report Writing & Finalization							✓	✓

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